

JEE Mains 2019 Chapter wise Question Bank

Ellipse – Questions

Q1

If tangents are drawn to the ellipse $x^2 + 2y^2 = 2$ at all points on the ellipse other than its four vertices then the mid points of the tangents intercepted between the coordinate axes lie on the curve :

- (1) $\frac{1}{4x^2} + \frac{1}{2y^2} = 1$ (2) $\frac{x^2}{4} + \frac{y^2}{2} = 1$
 (3) $\frac{1}{2x^2} + \frac{1}{4y^2} = 1$ (4) $\frac{x^2}{2} + \frac{y^2}{4} = 1$

11 Jan Morning

Q2

Let the length of the latus rectum of an ellipse with its major axis along x-axis and centre at the origin, be 8. If the distance between the foci of this ellipse is equal to the length of its minor axis, then which one of the following points lies on it?

- (1) $(4\sqrt{2}, 2\sqrt{2})$ (2) $(4\sqrt{3}, 2\sqrt{2})$
 (3) $(4\sqrt{3}, 2\sqrt{3})$ (4) $(4\sqrt{2}, 2\sqrt{3})$

11 Jan Evening

Q3

Let S and S' be the foci of an ellipse and B be any one of the extremities of its minor axis. If $\Delta S'BS$ is a right angled triangle with right angle at B and area $(\Delta S'BS) = 8$ sq. units, then the length of a latus rectum of the ellipse is :

- (1) 4 (2) $2\sqrt{2}$
 (3) $4\sqrt{2}$ (4) 2

12 Jan Evening

Q4

If the tangents on the ellipse $4x^2 + y^2 = 8$ at the points (1, 2) and (a, b) are perpendicular to each other, then a^2 is equal to :

- (1) $\frac{128}{17}$ (2) $\frac{64}{17}$ (3) $\frac{4}{17}$ (4) $\frac{2}{17}$

8 April Morning

Q5

In an ellipse, with centre at the origin, if the difference of the lengths of major axis and minor axis is 10 and one of the foci is at $(0, 5\sqrt{3})$, then the length of its latus rectum is:

- (1) 10 (2) 5 (3) 8 (4) 6

8 April Evening

Q6

If the line $x - 2y = 12$ is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

at the point $\left(3, \frac{-9}{2}\right)$, then the length of the latus rectum of the ellipse is :

- (1) 9 (2) $12\sqrt{2}$ (3) 5 (4) $8\sqrt{3}$

10 April Morning

Q7

The tangent and normal to the ellipse $3x^2 + 5y^2 = 32$ at the point P(2, 2) meet the x-axis at Q and R, respectively. Then the area (in sq. units) of the triangle PQR is :

- (1) $\frac{34}{15}$ (2) $\frac{14}{3}$ (3) $\frac{16}{3}$ (4) $\frac{68}{15}$

10 April Evening

Q8

If the normal to the ellipse $3x^2 + 4y^2 = 12$ at a point P on it is parallel to the line, $2x + y = 4$ and the tangent to the ellipse at P passes through Q (4,4) then PQ is equal to :

- (1) $\frac{5\sqrt{5}}{2}$ (2) $\frac{\sqrt{61}}{2}$ (3) $\frac{\sqrt{221}}{2}$ (4) $\frac{\sqrt{157}}{2}$

12 April Morning

Q9

An ellipse, with foci at $(0, 2)$ and $(0, -2)$ and minor axis of length 4, passes through which of the following points ?

- (1) $(\sqrt{2}, 2)$ (2) $(2, \sqrt{2})$
(3) $(2, 2\sqrt{2})$ (4) $(1, 2\sqrt{2})$

12 April Evening



mathongo

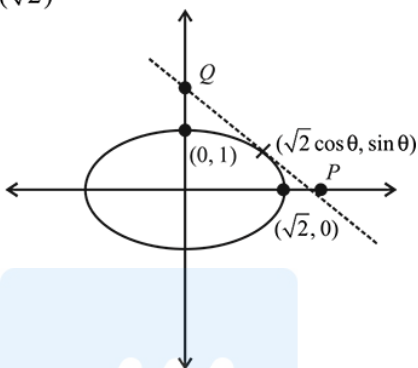
JEE Mains 2019 Chapter wise Question Bank

Ellipse - Solutions

Q1

(3) Given the equation of ellipse,

$$\frac{x^2}{(\sqrt{2})^2} + y^2 = 1$$



$$\frac{\sqrt{2} \cos \theta}{2} + y \sin \theta = 1$$

$$P\left(\frac{\sqrt{2}}{\cos \theta}, 0\right) \text{ and } Q\left(0, \frac{1}{\sin \theta}\right)$$

Let mid point be (h, k)

$$\Rightarrow h = \frac{1}{\sqrt{2} \cos \theta}, k = \frac{1}{2 \sin \theta}$$

$$\text{As } \cos^2 \theta + \sin^2 \theta = 1$$

$$\therefore \frac{1}{2h^2} + \frac{1}{4k^2} = 1$$

$$\text{Locus is } \frac{1}{2x^2} + \frac{1}{4y^2} = 1$$

11 Jan Morning

Q2

(2) Let the ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\text{Then, } \frac{2b^2}{a} = 8, 2ae = b^2 \text{ and } b^2 = a^2(1 - e^2)$$

$$\Rightarrow a = 8, b^2 = 32$$

Then, the equation of the ellipse

$$\frac{x^2}{64} + \frac{y^2}{32} = 1$$

Hence, the point $(4\sqrt{3}, 2\sqrt{2})$ lies on the ellipse.

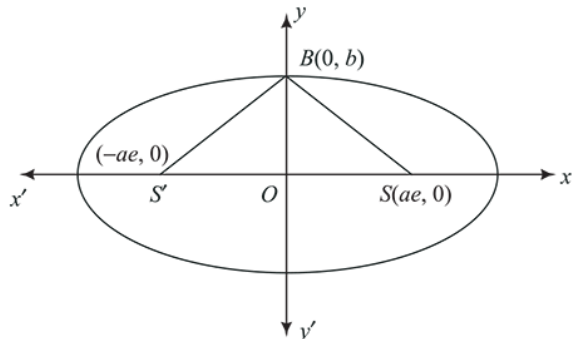
11 Jan Evening

Q3

Ellipse

(1) $\because \Delta S'BS$ is right angled triangle, then

$$(\text{Slope of } BS) \times (\text{Slope of } BS') = -1$$



$$\frac{b}{-ae} \times \frac{b}{ae} = -1$$

$$b^2 = a^2 e^2 \quad \dots(1)$$

Since, area of $\Delta S'BS = 8$

$$\Rightarrow \frac{1}{2} \cdot 2ae \cdot b = 8$$

$$b^2 = 8 \quad \dots(2)$$

From eqⁿ (1)

$$a^2 e^2 = 8$$

Also,
$$e^2 = 1 - \frac{b^2}{a^2}$$

$$\Rightarrow a^2 e^2 = a^2 - b^2$$

$$\Rightarrow 8 = a^2 - 8$$

$$\Rightarrow a^2 = 16$$

$$\text{Hence, required length of latus rectum} = \frac{2b^2}{a} = \frac{2(8)}{4}$$

$$= 4 \text{ units}$$

12 Jan Evening

Q4

JEE Mains 2019 Chapter wise Question Bank

(4) Since (a, b) touches the given ellipse $4x^2 + y^2 = 8$
 $\therefore 4a^2 + b^2 = 8 \quad \dots(1)$

Equation of tangent on the ellipse at the point $A(1, 2)$ is:

$$4x + 2y = 8 \Rightarrow 2x + y = 4 \Rightarrow y = -2x + 4$$

But, also equation of tangent at $P(a, b)$ is:

$$4ax + by = 8 \Rightarrow y = \frac{-4a}{b} + \frac{8}{b}$$

Since, tangents are perpendicular to each other.

$$\Rightarrow \frac{-4a}{b} = \frac{-1}{2}$$

$$\Rightarrow b = 8a \quad \dots(2)$$

from (1) & (2) we get:

$$\Rightarrow a = \pm \frac{2}{\sqrt{34}} \Rightarrow a^2 = \frac{2}{17}$$

8 April Morning

Q5

(2) Given that focus is $(0, 5\sqrt{3}) \Rightarrow |b| > |a|$

Let $b > a > 0$ and foci is $(0, \pm be)$

$$\therefore a^2 = b^2 - b^2 e^2 \Rightarrow b^2 e^2 = b^2 - a^2$$

$$be = \sqrt{b^2 - a^2} \Rightarrow b^2 - a^2 = 75 \quad \dots(i)$$

$$\therefore 2b - 2a = 10 \Rightarrow b - a = 5 \quad \dots(ii)$$

From (i) and (ii)

$$b + a = 15 \quad \dots(iii)$$

On solving (ii) and (iii), we get

$$\Rightarrow b = 10, a = 5$$

$$\text{Now, length of latus rectum} = \frac{2a^2}{b} = \frac{50}{10} = 5$$

8 April Evening

Q6

Ellipse

- (1) Equation of tangent to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at $\left(3, -\frac{9}{2}\right)$ is,

$$\frac{3x}{a^2} - \frac{9y}{2b^2} = 1$$

But given equation of tangent is, $x - 2y = 12$

$$\therefore \frac{3}{a^2} = \frac{-9}{2b^2 \cdot (-2)} = \frac{1}{12} \quad (\text{On comparing})$$

$$\Rightarrow a^2 = 3 \times 12 \text{ and } b^2 = \frac{9 \times 12}{4}$$

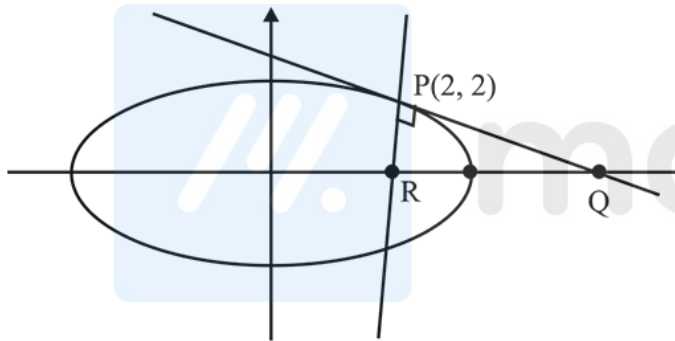
$$\Rightarrow a = 6 \text{ and } b = 3\sqrt{3}$$

$$\text{Therefore, latus rectum} = \frac{2b^2}{a} = \frac{2 \times 27}{6} = 9$$

10 April Morning

Q7

$$(4) 3x^2 + 5y^2 = 32 \Rightarrow \frac{3x^2}{32} + \frac{5y^2}{32} = 1$$



Tangent on the ellipse at P is

$$\frac{3(2)x}{32} + \frac{5(2)y}{32} = 1 \Rightarrow \frac{3x}{16} + \frac{5y}{16} = 1$$

$$\therefore \text{co-ordinates of Q will be } \left(\frac{16}{3}, 0\right)$$

$$\text{Now, normal at P is } \frac{32}{3(2)} - \frac{32y}{5(2)} = \frac{32}{3} - \frac{32}{5}$$

$$\therefore \text{co-ordinates of R will be } \left(\frac{4}{5}, 0\right)$$

$$\text{Hence, area of } \Delta PQR = \frac{1}{2} (PQ)(PR)$$

$$= \frac{1}{2} \sqrt{\frac{136}{9}} \cdot \sqrt{\frac{136}{25}} = \frac{68}{15}$$

10 April Evening

JEE Mains 2019 Chapter wise Question Bank

Q8

- (1) Slope of tangent on the line $2x + y = 4$ at point P is $\frac{1}{2}$.

Given ellipse is,

$$3x^2 + 4y^2 = 12 \Rightarrow \frac{x^2}{2^2} + \frac{y^2}{(\sqrt{3})^2} = 1$$

Let point $P(2\cos \theta, \sqrt{3} \sin \theta)$

$\therefore$ equation of tangent on the ellipse, at P is,

$$\frac{x}{2} \cos \theta + \frac{y}{\sqrt{3}} \sin \theta = 1$$

$$\Rightarrow m_T = -\frac{\sqrt{3}}{2} \cot \theta$$

$$\therefore \text{both the tangents are parallel} \Rightarrow -\frac{\sqrt{3}}{2} \cot \theta = \frac{1}{2}$$

$$\Rightarrow \tan \theta = -\sqrt{3} \Rightarrow \theta = \pi - \frac{\pi}{3} \text{ or } \theta = 2\pi - \frac{\pi}{3}$$

$$\text{Case-1: } \theta = \frac{2\pi}{3}, \text{ then point } P\left(-1, \frac{3}{2}\right) \text{ and } PQ = \frac{5\sqrt{5}}{2}$$

$$\text{Case-2: } \theta = \frac{5\pi}{3}, \text{ then tangent does not pass through } Q(4, 4).$$

12 April Morning

Q9

- (1) Let the equation of ellipse :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Given that length of minor axis is 4 i.e. $a = 4$.

Also given $b = 2$

$$\therefore a^2 = b^2(1 - e^2) \Rightarrow 4 = b^2 - 4 \Rightarrow b = 2\sqrt{2}$$

$$\text{Hence, equation of ellipse will be } \frac{x^2}{4} + \frac{y^2}{8} = 1$$

$$\therefore (\sqrt{2}, 2) \text{ satisfies this equation.}$$

$$\therefore \text{ellipse passes through } (\sqrt{2}, 2).$$

12 April Evening